

Software Engineering

Spring 2026
Lab 4





Team Mamber

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01

Introduction

Problem Statement

- Define your problem here

Finding a parking space in crowded areas is often difficult and time-consuming. Drivers usually spend a lot of time searching for available parking spots, which leads to traffic congestion, fuel waste, and frustration. Traditional parking systems do not provide real-time information about available parking spaces, making the process inefficient and inconvenient for drivers.



Proposed System

- Define how you solve the problem you choose

The proposed system is a Smart Parking System that allows drivers to view available parking slots and reserve them before arriving at the parking area. The system provides real-time information about parking availability and helps users book or cancel parking slots easily through a simple interface. Administrators can manage parking spaces by adding, updating, or removing slots. This system aims to reduce traffic congestion, save time for drivers, and improve parking management efficiency.





02

Requirement Analysis

Functional Requirements

- Type system requirements
 - User Registration
 - User Login
 - View Available Parking Slots
 - Book Parking Slot
 - Cancel Booking
 - View Booking History
 - Add Parking Slot
 - Update Parking Slot
 - Delete Parking Slot
 - View All Bookings



Non-Functional Requirements

- Type system Non-Functional requirements
 - Performance
 - Security
 - Usability
 - Availability
 - Reliability
 - Scalability
 - Maintainability
 - Compatibility

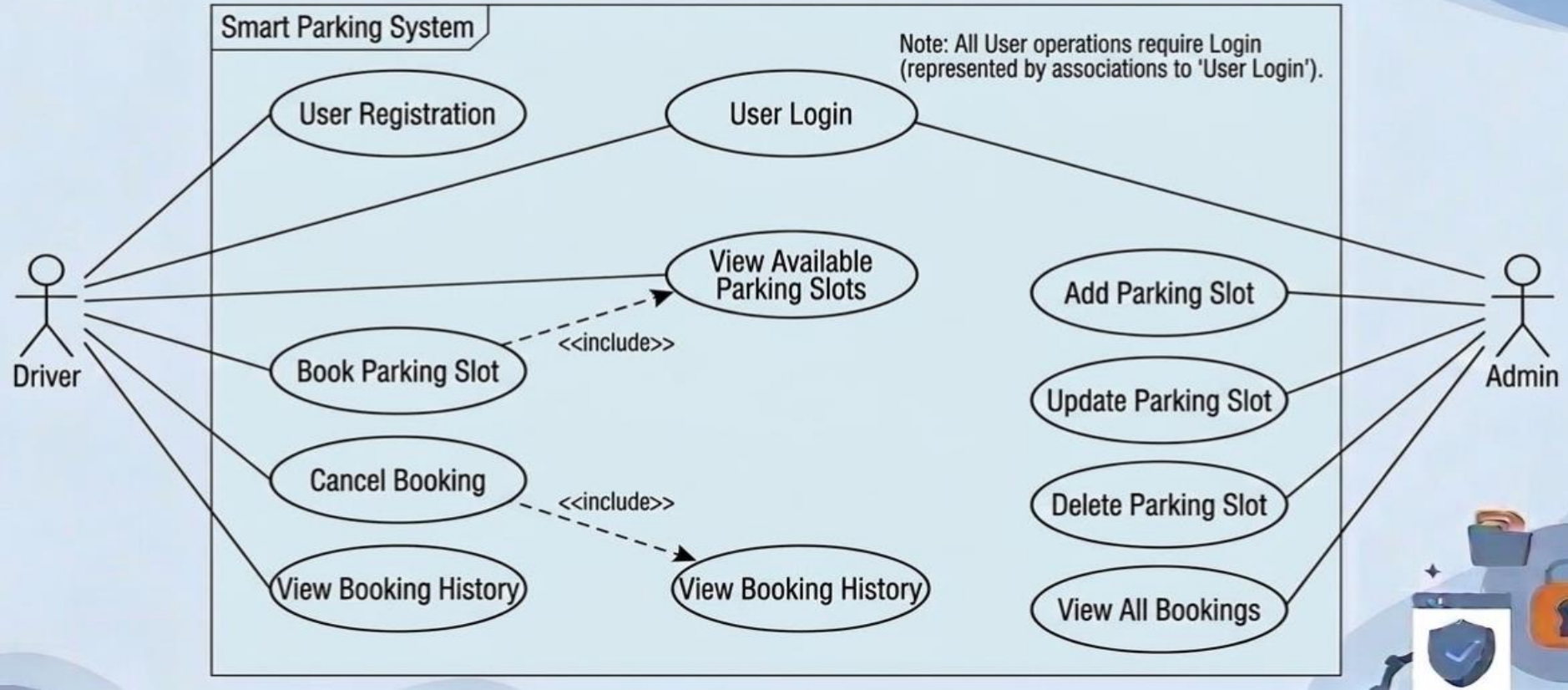




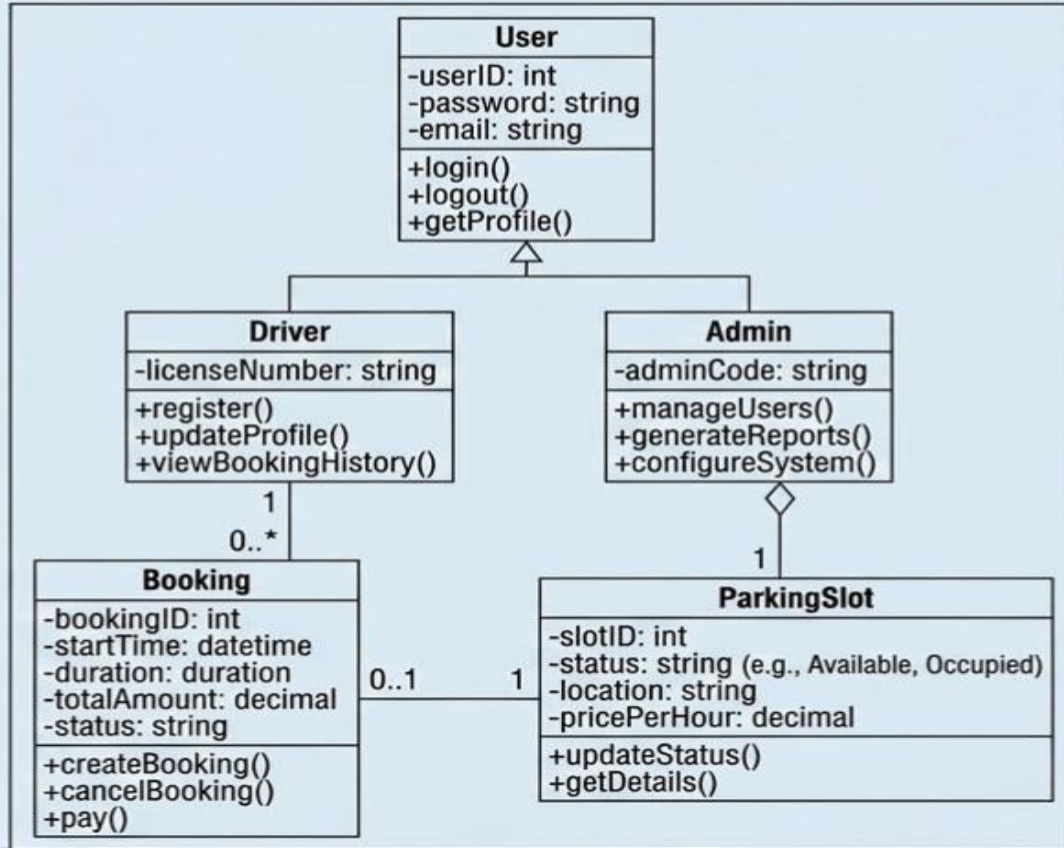
03

UML Diagrams

Use Case Diagram

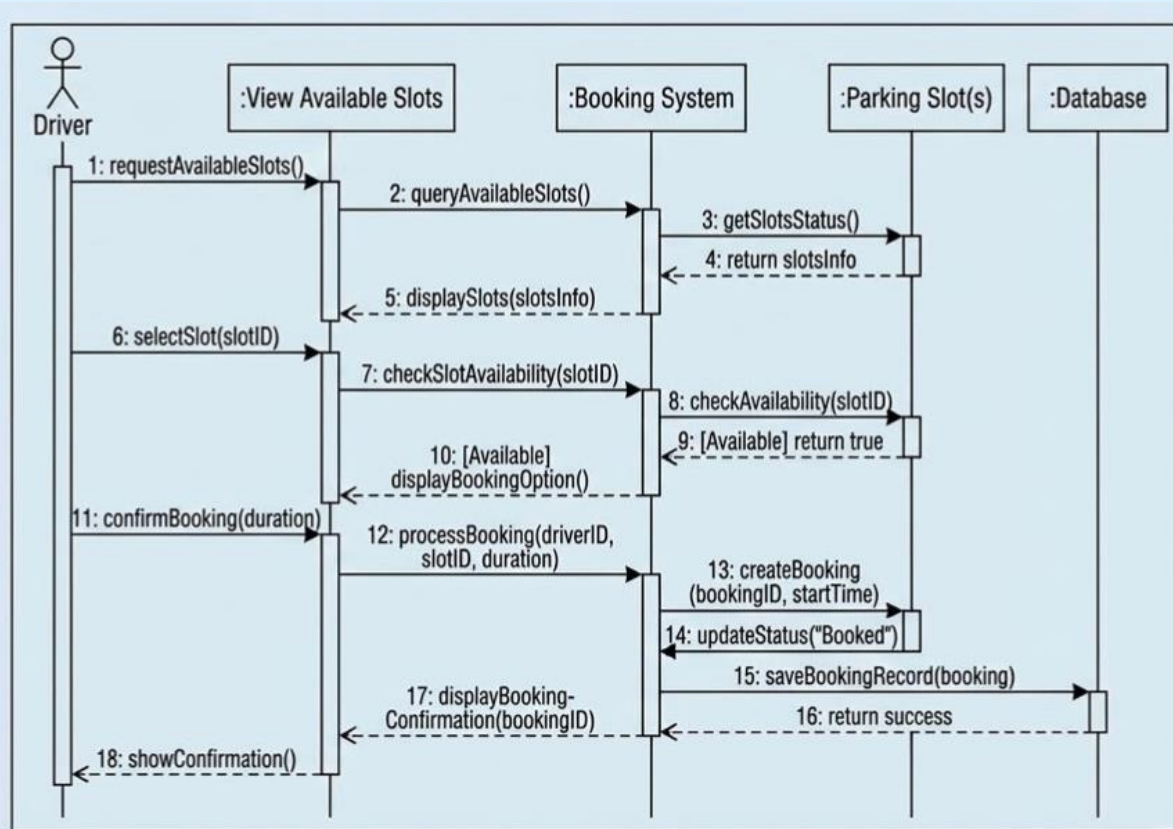


Class Diagram



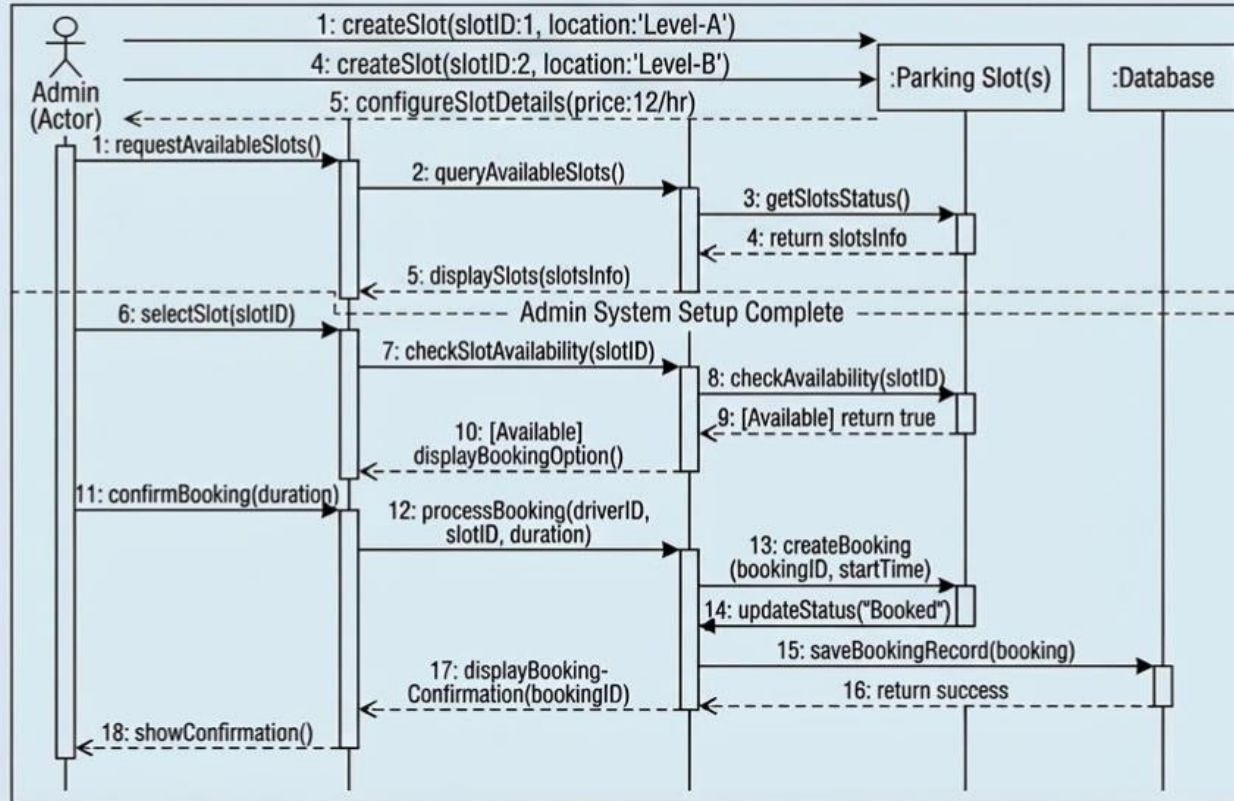
Sequence Diagram

USER



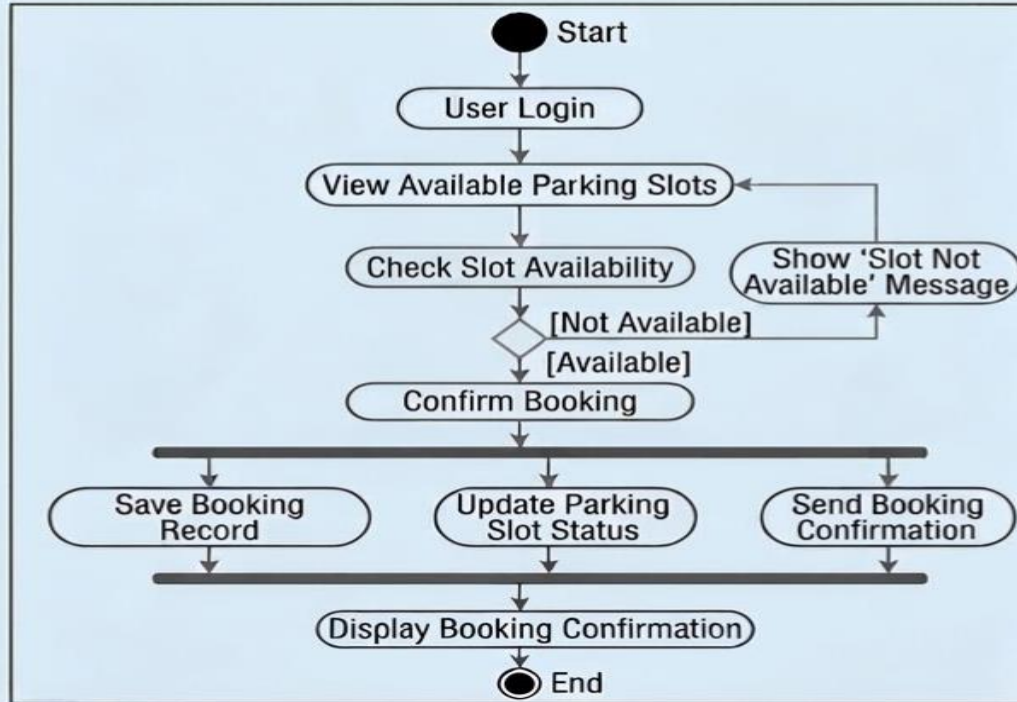
Sequence Diagram

ADMIN



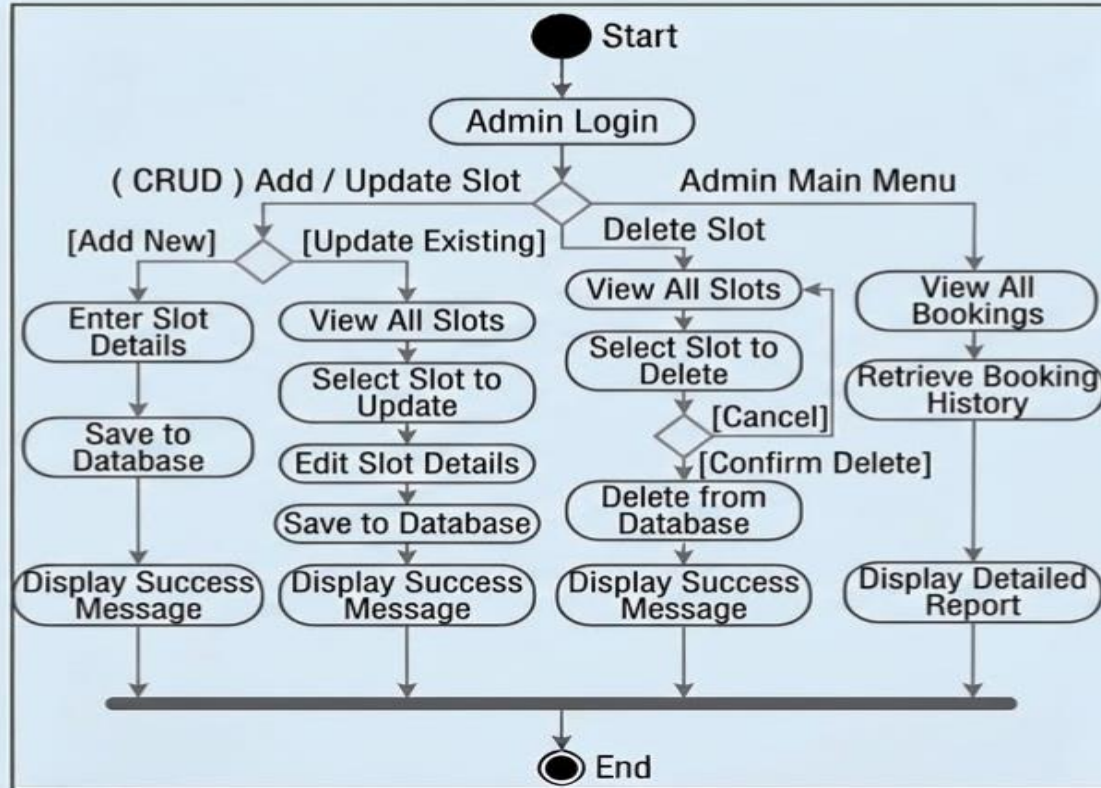
Activity Diagram

USER



Activity Diagram

ADMIN



THANKS!

